NeuroBridge-S4 Supporting Documentation

Graph-informed BACI, sensitivity, ablation, and HRP decision-support logic

What HRP Gets

Output	Meaning for HRP
Individual Adaptation Profile	Subject-specific map of shifted domains, coherence, recovery status, monitoring priority, and next data need.
Crew-Level Summary	Shared and divergent patterns across four people without false group statistics.
Graph Pack	Knowledge graph, observed proxy coherence graph, individual adaptation networks, decision graph.
BACI Sensitivity	Shows whether coherence ranking is robust to threshold changes.
Ablation View	Shows why domain compression, BACI, and graphs add value beyond raw z-scores.
Countermeasure Priority Map	Links patterns to sleep/circadian, immune, cognitive, psychological, workload, recovery, and nutrition/hydration review categories.

Rubric Alignment

Rubric area	NeuroBridge-S4 response
Methodological rigor	Control Ladder, within-subject logic, baseline imbalance, missingness awareness, graph separation, BACI sensitivity.
Demonstration quality	Executable notebook, real NHANES participants, saved figures/tables, reviewer-friendly narrative and outputs.
Artemis applicability	Direct mapping to Standard Measures, ARCHeR, Immune Biomarkers, and mission phases.
Innovation and impact	Graph-informed biological coherence as a bridge from sparse data to adaptation management.

Graph Governance Rules

• Every graph is labeled by type: conceptual knowledge, data-derived observed coherence, or decision translation.

• Conceptual edges are not presented as findings from NHANES.

• Observed edges are thresholded and reproducible.

• Decision edges are presented as monitoring priorities and countermeasure considerations, not medical advice.

BACI Formula - Demonstration Version

For the executable notebook, BACI is intentionally transparent. It uses domain-level deviations and tests multiple thresholds. A full Artemis version would add longitudinal alignment, recovery delay, graph centrality, and validated self-report weighting.

Term	Meaning
ActiveDomainCount(threshold)	Number of biological domains above the deviation threshold.
MeanActiveMagnitude	Average magnitude among active domains.
BreadthFactor	Fraction of measured domains involved.
GraphSupport	Optional multiplier if active domains are connected in the knowledge/observed graph.
BACI 0-100	Normalized coherence score for review prioritization.

Ablation Logic

Layer	Without it	With it
Raw variables	Scattered biomarkers; hard to interpret.	Reference deviation shows what is unusual.
Biological domains	Too many columns.	System-level interpretation.
BACI	No way to rank multi-domain convergence.	Coherence ranking for monitoring priority.
Graphs	Flat table; weak explanatory bridge.	Relational structure of adaptation is visible.
Decision map	Academic analysis only.	Operational monitoring and support priorities.

Biochemistry - Neurobiology - Cognition - Psychology Bridge

The project explicitly avoids simplistic claims like biomarker X equals psychological state Y. Instead it models acute functional states as convergent patterns across interacting systems. Sleep/circadian disruption, autonomic dysregulation, HPA-axis stress biology, immune-inflammatory signaling, cognitive fatigue, and validated self-report can form a graph-supported vulnerability context relevant to psychological functioning and mission performance.

Functional layer	Observable evidence	Management interpretation
Recovery capacity	Sleep/activity, HRV/autonomic measures, fatigue, return-to-baseline.	Recovery protocol and workload pacing.
Stress-immune coupling	Stress hormones, immune markers, latent virus signals, inflammation.	Immune follow-up and stress-load monitoring.
Cognitive load / fatigue	Cognitive tasks, sleep disruption, fatigue surveys, autonomic strain.	Cognitive monitoring and task pacing.
Acute cognitive-perceptual vulnerability	Sleep + autonomic + fatigue + self-report + stress context.	Psychological check-in and sensory/workload context review.

NeuroBridge-S4 is not a model of astronauts. It is a model of how to reason responsibly about astronauts when the data are rare, fragmented, biologically rich, and operationally consequential.